

Math 290

Practice Quiz 2 — Version F

(extra practice, Sections 6–11)

1. Short response.

(a) Evaluate the following proposed theorem and proof.

Theorem: Let $a, b, x \in \mathbb{R}$ with $b \geq 0$. If $|x - a| \leq b$, then $a - b \leq x \leq a + b$.

Proof. Suppose that $|x - a| \leq b$. Since $|x - a| = x - a$, we have $0 \leq x - a \leq b$. Adding a , we find that $a \leq x \leq a + b$. Since $a - b \leq a$, we see that $a - b \leq x \leq a + b$. $\square$

- A. The theorem is true and the proof is correct.
- B. The theorem is false.
- C. The theorem is true, but the proof is incorrect because it assumes that $b \geq 0$.
- D. The theorem is true, but the proof is incorrect because it does not deal with the case where $x - a < 0$.
- E. The theorem is true, but the proof is incorrect because you cannot add a to every part of an inequality.

(b) True or false: To disprove the statement “There exists $x \in \mathbb{R}$ such that $x^2 < -1$,” it is enough to give one value of x with $x^2 \geq -1$.

(c) True or false: $(28, 6) \in \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x \equiv y \pmod{11}\}$.

2. Let $x \in \mathbb{Z}$. Prove that $5x - 1$ is even only if x is odd. (Be careful to prove the correct implication.)
3. Prove that for every $n \in \mathbb{Z}$, there exists a *unique* $m \in \mathbb{Z}$ such that $2m \leq n < 2m + 2$.
4. Fix X to be the set of integers that are congruent to -1 modulo 6, and fix Y to be the set of integers that are congruent to 2 modulo 3. Prove that $X \subseteq Y$, and then prove that $X \neq Y$.